Rahul Muttanawar

AI Engineer

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PROFESSIONAL SUMMARY

Innovative **AI/ML Engineer** with **2+ years experience** shipping **LLM/RAG** systems in production. I assume **end-to-end, DRI-level ownership** of AI microservices and data pipelines—accountable for **architecture**, model development, **API design**, **CI/CD**, **SLA/SLO-based observability**, and incident command with postmortems. Built **NLP pipelines**, **embedding retrieval**, and **RAG** over vector DBs (Milvus). Proficient in **Python, FastAPI, AI Agent ,TensorFlow, PyTorch , Langgraph , Agentic AI , MCP** ; experienced with **MLflow, Docker**, **Apache Airflow**, and **GPU inference** (llama.cpp). Deliver scalable AI across **on-premises** and **cloud-native** (Azure), AWS environments.

EXPERIENCE

Genzeon Technology Solutions, Pune, Maharashtra May 2023 – Present

AI ML Engineer –GenAI

AI-Powered Document Processing for Claims Automation

- Built **ML-based document classification** and **key data extraction** pipelines using Small Language Models (SLMs), improving claims automation and data accuracy.
- Developed and deployed text summarization and classification models using Hugging Face Transformers and FastAPI, serving 10K+ requests per month.
- Developed scalable **FastAPI microservices** for document ingestion and retrieval using asynchronous processing and dependency injection, improving system performance.

AI & Chatbot Development

- Implemented **RAG-based chatbots** over structured and unstructured data (PDFs, CCDA documents) utilizing **Milvus** to streamline user interactions, reducing manual lookup by **40%**.
- Leveraged generative AI techniques, **including fine-tuning LLMs** (e.g., GPT variants) and implementing **prompt engineering strategies**, to enhance chatbot contextual understanding and **response generation by 25%**.
- Conducted **A/B testing and evaluation** to ensure fairness and accuracy across user cohorts.

Workflow and DevOps Automation

- Automated **API workflows** with large-scale data pipeline using **Apache Airflow**, enabling **efficient orchestration, cleaning/tokenization of 100M+ records**, monitoring, and error handling.
- Streamlined API integrations by implementing **event-driven automation**, improving system reliability and reducing manual intervention.
- Developed and deployed **systemd services** for each application on VM environments, ensuring reliable service management.

Centralized Logging for Claim Lifecycle Monitoring

- Built a centralized logging pipeline using **Filebeat, Kafka, Logstash,** and **Elasticsearch** to monitor claim processing across microservices in real time.
- Designed **Kibana dashboards** to visualize the complete claim lifecycle, improving traceability and reducing debugging time by **60%**.
- Integrated **Prometheus** with custom metrics to monitor model performance in production, reducing inference latency by **40%**.

PROJECTS

Custom YOLO Model Training *Python, YOLOv5, DocTR, Azure Form Recognizer*

- Developed custom-trained **YOLOv5** models for checkbox and signature detection, improving document-processing accuracy by **80%**.
- Optimized detection/OCR workflow by integrating **DocTR** and **Azure Form Recognizer**, reducing OCR costs by **75%**.

Healthcare SLM Integration *LangChain, OpenAI API, FastAPI*

- Built an **SLM service** specialized for medical domains (e.g., cardiology, neurology), increasing data-retrieval precision by **40%**.
- Unified multiple data sources (documents, URLs) to enable seamless retrieval and conversation over healthcare records.

Performance Benchmarking of SLMs/LLMs on GPUs *RTX 5090, H100, PyTorch, llama.cpp*

- Conducted extensive **GPU inference benchmarking** for SLMs and LLMs (batching, quantization, threading), optimizing **p95 latency** and throughput on **RTX 5090/H100**.

SKILLS

Languages:	Python, SQL, Bash, Javascript
ML & AI:	RAG, Embeddings, Transformers, Prompt Engineering, NLP, Langgraph,Agentic AI , LangChain
LLM Runtimes:	llama.cpp, Model Fine-tuning (LoRA/QLoRa)
Frameworks:	PyTorch, TensorFlow, scikit-learn, FastAPI, Pandas, NumPy, OpenCV, REST, pytest
Databases:	PostgreSQL, Milvus, Elasticsearch
Orchestration:	Apache Airflow, NGINX, Redis
Observability:	Kafka, Logstash, Filebeat, Kibana
Monitoring:	Telegraf, InfluxDB, Chronograf
Platforms:	Azure, Linux, RHEL, Windows, NVIDIA GPUs (4090, 5090)

CERTIFICATIONS

- Microsoft Azure Fundamentals (**AZ-900**) - Completed 2023
- Microsoft AI Fundamentals (**AI-900**) - Completed 2023

AWARDS AND HONORS

- **Star Achievement Award**, Genzeon Technology Solutions – July 2025. Awarded for exceptional performance, dedication, and contributions to the company's success, presented by **CEO Ilanko Kumaresan**.
- Innovation of the Quarter for **developing On-Premise solution.**
- Team Recognition for Onboarding a client and **building a product from scratch**, Genzeon Technology Solutions – November 2024. Presented by **Senior Director, Practice and Delivery**.

EDUCATION

- **Post Graduate Diploma in Advanced Computing (PG-DAC)** 2022 - 2023
  CDAC , ACTS , Pune, Maharashtra
- **Bachelor of Engineering -** 2018 - 2021
  D Y Patil College of Engineering, Kolhapur, Maharashtra